SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1516

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**
Question Count: 5

Code of Conduct

I _____VINIL KUMAR .S (192511226)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: _____

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

QUESTION 1: Type-1 and Type-2 Hypervisors for Mission-Critical Applications

Introduction

A hypervisor creates and manages virtual machines (VMs) on a physical server. For a cloud datacenter hosting mission-critical applications, the hypervisor must provide reliable performance, strong isolation, efficient resource allocation and manageable operations. Type-1 and Type-2 hypervisors differ mainly in where they run.

Type-1 and Type-2 Architecture

Type-1 Hypervisor	Type-2 Hypervisor
Runs directly on physical hardware.	Runs as software on a host operating system.
VMs access hardware through the hypervisor.	VMs access hardware through the hypervisor and host OS.

Performance Comparison

A Type-1 hypervisor has fewer software layers between the VM and hardware. This generally reduces overhead and improves the predictability of CPU scheduling, memory management, storage access and network performance. A Type-2 hypervisor depends on the host operating system, whose background tasks and resource demands may add latency. Therefore, Type-1 is usually more suitable for high-demand production workloads.

Security and Isolation

Type-1 hypervisors have a smaller software stack and are designed for server virtualization, but a hypervisor vulnerability can still affect multiple VMs. Type-2 hypervisors also depend on the security of the host OS; if it is compromised, hosted VMs may be exposed. Both types require updates, access control, secure configuration and monitoring. In a production datacenter, the reduced dependency on a general-purpose host OS makes Type-1 the stronger choice in general.

Manageability and Availability

Type-1 platforms commonly support centralized VM management, resource allocation, monitoring, live migration and high-availability features. These functions help administrators maintain service continuity and respond to changing demand. Type-2 platforms are often easier to use for desktop testing and development but are less suited to large-scale production management.

Summary and Recommendation

Criterion	Type-1	Type-2
Performance	Lower overhead; predictable for server workloads.	Extra host OS may add overhead.
Security	Fewer software layers; requires hypervisor hardening.	Host OS adds another attack surface.
Management	Centralized datacenter features.	Often suited to local development/testing.
Best use	Production and mission-critical cloud services.	Labs, desktops and application testing.

Recommendation: Select a Type-1 hypervisor for mission-critical applications. Support it with role-based access control, timely patching, redundant hardware, monitoring, backups and tested disaster recovery. Type-2 is better reserved for development and testing.

QUESTION 2: Isolation of Finance, Healthcare and Student Workloads

Problem Interpretation

The cloud provider must host finance, healthcare and student-portal workloads on the same physical hardware while preventing unauthorized access between tenants. Since these workloads may contain sensitive information, isolation must cover compute, memory, storage, networking and administration.

Role of Virtualization in Isolation

The provider can place each workload in a separate VM, with its own virtual disks, virtual network interfaces and access policies. The hypervisor separates VM execution and controls access to physical CPU and memory. Virtual switches, firewall rules and network segmentation restrict traffic between workloads. Separate storage permissions and encryption keys reduce the risk of one tenant reading another tenant's data.

Finance VM | Healthcare VM | Student Portal VM
Separate virtual disks, virtual NICs and security policies
Hypervisor enforces VM boundaries and allocates shared resources
Shared physical server: CPU + RAM + storage + network

Attack Surface 1: Hypervisor and VM Escape

A flaw in the hypervisor or a virtual device may allow an attacker inside one VM to access the host or another VM. This is dangerous because several tenants share the same physical server. Reduce the risk by promptly patching the hypervisor, disabling unnecessary virtual devices, applying hardened configurations, limiting administrative access and monitoring suspicious VM activity.

Attack Surface 2: Virtual Networks and Management Interfaces

Incorrect virtual-switch rules, exposed management APIs, weak passwords or excessive administrator privileges can enable unauthorized access or traffic interception. Use network segmentation, firewalls, multi-factor authentication (MFA), least-privilege role-based access control, encrypted management connections and audit logs. Management interfaces should not be exposed directly to public networks.

Additional Safeguards

- Encrypt sensitive data both at rest and in transit.
- Apply separate tenant permissions and rotate keys securely.
- Set CPU, memory and I/O limits to reduce noisy-neighbour effects.
- Conduct vulnerability assessments and test isolation boundaries.
- Maintain protected backups and an incident-response plan.

Conclusion

Virtualization provides logical separation on shared hardware, but isolation is effective only when the hypervisor, network, storage and management plane are configured securely. The two major attack surfaces are the hypervisor/VM boundary and the virtual network/management interfaces. Layered controls and regular testing are required to protect all three workloads.

QUESTION 3: VM Host Utilization and Corrective Actions

3.1 Utilization Snapshot

Metric	Observed value
CPU utilization	92%
Memory utilization	88%
Storage I/O latency	35 ms
Average VM boot time	170 seconds

3.2 CPU Bottleneck

CPU utilization of 92% is very high and leaves little spare capacity for demand spikes. VMs may wait longer for CPU time, increasing application response time. Administrators should identify CPU-heavy VMs and processes, right-size vCPU allocations, rebalance workloads across hosts and add capacity if high usage persists.

3.3 Memory Bottleneck

Memory utilization of 88% indicates limited headroom. If physical memory becomes insufficient, the host may reclaim memory or swap data to disk, which can significantly reduce performance. Check host and guest memory pressure, remove unnecessary reservations, right-size VM memory and add RAM or migrate workloads when measurements justify it.

3.4 Storage I/O Bottleneck

Storage I/O latency of 35 ms suggests that read or write requests are taking a relatively long time. Possible causes include datastore contention, high queue depth, insufficient IOPS or slow storage devices. Review throughput, IOPS and queue metrics; distribute disk-intensive VMs, optimize storage policies and move critical workloads to faster storage where appropriate.

3.5 Slow VM Boot Time

An average boot time of 170 seconds may result from storage delays, CPU or memory contention, heavy startup services, oversized VM images or many VMs starting simultaneously. Measure each boot stage, remove unnecessary startup tasks, optimize images and stagger scheduled starts. Improving storage performance may also reduce boot delays.

3.6 Prioritized Corrective Plan

Priority	Action	Reason
1	Collect time-series host, VM and datastore metrics.	Confirm whether the problem is persistent and identify the main consumers.
2	Right-size VMs and rebalance workloads.	Reduce CPU, memory and I/O contention.
3	Optimize storage and stagger boot activity.	Reduce I/O queues and boot-time spikes.
4	Add host, RAM or storage capacity if needed.	Provide headroom when optimization is insufficient.

3.7 Conclusion

The snapshot points to possible resource contention: CPU and memory usage are high, storage latency is elevated, and VM startup is slow. These symptoms may be related, but the exact cause must be verified with performance metrics and boot logs before purchasing additional hardware.

QUESTION 4: Software-Defined Datacenter Architecture

4.1 Introduction

A Software-Defined Datacenter (SDDC) uses software to abstract and manage compute, storage and networking resources. It replaces many manual, device-by-device tasks with centralized policies and automation, making infrastructure faster to provision and easier to scale.

4.2 Compute Virtualization

Compute virtualization divides physical server resources among VMs or workloads. In an SDDC, templates and orchestration tools can provision VMs quickly, adjust resource allocations and move workloads when demand changes. This improves utilization and reduces the time needed to deliver application environments.

4.3 Storage Virtualization

Storage virtualization combines capacity from multiple devices into software-managed pools. Policies can define performance, capacity, replication and data-protection needs. Administrators can allocate storage more consistently and scale capacity without managing every device separately.

4.4 Network Virtualization

Network virtualization creates logical networks, virtual switches and software-defined routing over physical infrastructure. Networks and security policies can be deployed centrally, enabling faster environment creation and consistent segmentation. This also helps isolate applications and tenants.

4.5 Architecture Overview

Applications and tenant workloads
SDDC control plane: orchestration + automation + policies + monitoring
Virtualized resource pools: compute + storage + network
Physical servers + storage devices + network equipment

4.6 Agility Compared with a Traditional Datacenter

Traditional Datacenter	**SDDC**
Manual configuration across separate devices.	Centralized software control and repeatable automation.
Provisioning may take longer and require several teams.	Templates and orchestration speed up provisioning.
Scaling may require manual changes to individual systems.	Pooled resources can be reallocated based on policy.
Tools and monitoring may be separated by infrastructure type.	Central monitoring improves visibility and consistency.

4.7 Security and Limitations

The SDDC control plane is a high-value target because it can manage many resources. Protect it with MFA, least-privilege permissions, secure APIs, audit logs, backups and high availability. Test automation before deployment and monitor capacity, compatibility and licensing.

4.8 Conclusion

SDDC improves agility by automating compute provisioning, pooling storage and controlling networks through software. It enables faster scaling and more consistent policies than traditional manual operations, provided that the control plane and automation workflows are secured.

QUESTION 5: Secure Identity and Privacy in Multi-Cloud Federation

5.1 Problem Analysis

A multi-cloud federation allows services from different cloud providers to trust a common identity arrangement. The project failed because providers used inconsistent usernames, identifiers, group names, role definitions or token formats. As a result, users may be denied access incorrectly or receive excessive permissions.

5.2 Authentication and Authorization

Authentication confirms who the user is; authorization determines what the user can do. Federation must handle both correctly. Providers should validate the identity issuer and token, then map the user's attributes to local permissions. A token should not be trusted merely because it is present.

5.3 Use Standard Federation Protocols

Use a trusted Identity Provider (IdP) or federation broker and an established protocol such as OpenID Connect (OIDC) or SAML 2.0, depending on provider support. Register each cloud as a trusted relying party or service provider, and document which identities and claims it accepts.

5.4 Normalize Identities and Roles

Create a shared schema for subject identifiers, tenant IDs, groups and roles. Prefer stable, non-reassigned identifiers over usernames or email addresses alone. Define explicit mappings between each provider's roles and the common schema. If a required attribute is missing or ambiguous, deny access until it is corrected.

5.5 Secure Tokens and Access

- Validate token signature, issuer, audience, expiry and required scope.
- Use MFA for sensitive access and apply least-privilege RBAC or ABAC.
- Protect signing keys, rotate them safely and use encrypted connections.
- Use short-lived tokens where practical and provide revocation/session termination.
- Separate administrator accounts and review privileged permissions regularly.

5.6 Privacy Protection

Share only the attributes needed by the receiving service. Avoid disclosing sensitive financial, health or personal information without a valid need and authorization. Use pseudonymous identifiers where feasible, encrypt data in transit and at rest, set retention limits, protect logs and follow applicable data-residency and privacy requirements.

5.7 Monitoring and Federation Flow

User authenticates with the trusted Identity Provider
IdP issues a signed, time-limited token or assertion
Cloud provider validates token and maps attributes to local roles
Policy grants least-privilege access; security events are logged

Monitor unusual sign-ins, role changes and failed validations. Test certificate expiry, revoked accounts, provider outages and compromised credentials before deployment.

5.8 Conclusion

A secure federation requires consistent identity attributes, explicit trust, strong token validation and least-privilege authorization. Privacy is maintained by minimizing shared data, encrypting information and limiting retention. All participating providers should agree on mappings and test the design before production use.